

Lunar-KAG: Knowledge Augmented Generation for Scientific Discovery in Lunar Exploration

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Abstract

Retrieval augmented generation (RAG) has become a fundamental framework for knowledge-intensive question answering. However, the existing RAG and Knowledge graph-enhanced methods operate at fragment-level or triplet-level granularity, which may not align with the inherently n -ary structure of lunar scientific knowledge. To address this, we propose **Lunar-KAG**, a hypergraph-enhanced RAG framework that models scientific discovery as an entity-hyperedge bipartite structure, preserving n -ary event units. Lunar-KAG introduces a dual-path dense retrieval mechanism that simultaneously matches queries in entity and hyperedge semantic spaces, respectively. The retrieved seeds are then organized into a query-specific local sub-hypergraph, providing structurally consistent evidence for downstream generation. Experiments on SpaceQA, CosmoPaperQA, and LunarRQA show that Lunar-KAG consistently improves both retrieval quality and generation reliability over graph-based baselines without increasing inference cost.

CCS Concepts

• **Applied computing** → **Physical sciences and engineering**; • **Information systems** → **Information retrieval**.

Keywords

Retrieval-Augmented Generation, Information Retrieval, Natural Language Processing, Question Answering, Lunar Exploration

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1 Introduction

The rapid development of LLMs establishes a new era for scientific discovery, enabling AI systems to autonomously execute research workflows and formulate insights [22, 36, 38]. In space science, valuable discovery requires integrating highly multi-sourced inputs, encompassing extensive literature corpora, numerical simulations, and large-scale observational datasets. However, their deployment in this specialized research is constrained by static knowledge and the inherent knowledge cut-off during the post-training [17]. This limitation prevents the integration of emerging mission description and recent publications, resulting in ungrounded responses [19].

RAG mitigates these issues by grounding generation in external scientific corpora [9]. Standard RAG employs text chunks as basic retrieval units via sparse [37] or dense matching [4, 23]. While effective for local fact extraction, chunk-level retrieval fails to recover complete event structures for cross-document, multi-entity queries. Graph-based RAG addresses this by introducing knowledge graphs, parsing text into entity-relation triplets for structured retrieval [25]. However, scientific facts inherently manifest as n -ary events. Decomposing these concurrent multi-entity associations into binary edges disrupts their original structural integrity. Hypergraph mitigate this by modeling relations as hyperedges to preserve high-order topologies. Existing methods like HyperGraphRAG [21] and Hyper-RAG [7] rely on a single semantic space or a serial retriever, aligning queries with discrete entities prior to relational expansion.

In this paper, we present Lunar-KAG, a hypergraph-guide RAG framework, formalizes scientific literature as an entity-hyperedge bipartite structure. Entity nodes represent discrete scientific concepts; hyperedges encode n -ary event-level relations. A dual-path retrieval scheme operates across decoupled entity and hyperedge semantic spaces. Dense matching executes independently in both spaces per query, generating concept-level and event-level retrieval seeds without reducing events to binary relations. Integration of retrieved entities and hyperedges via the underlying incidence structure induces a query-specific local sub-hypergraph. This configuration preserves higher-order relational units and structural dependencies, maintaining event-level integrity for consistent multi-entity evidence aggregation during generation. Overall, the main contributions are summarized as follows:

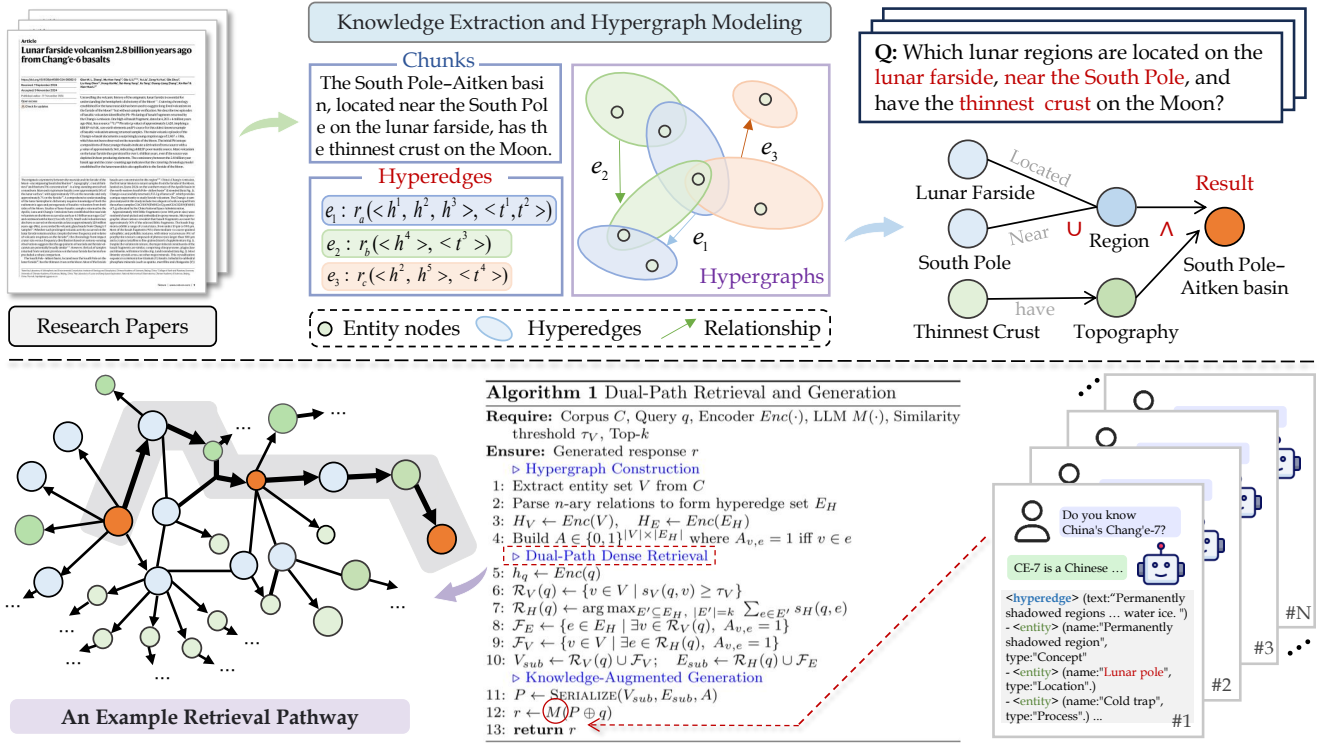


Figure 1: The overall framework of the Lunar-KAG.

- **(Framework)** We propose Lunar-KAG (Figure 1), which models information as an entity-hyperedge bipartite graph.
- **(Retrieval)** We design a dual-path retrieval module that employs topological expansion, guided by an incidence matrix, to aggregate structurally consistent evidence.
- **(Performance)** On open-domain benchmarks, Lunar-KAG improves MRR@10 and faithfulness while reducing inference latency compared with GraphRAG baselines.

2 Related Work

Recent progress in information retrieval has accelerated the adoption of LLMs in deep-space exploration. Lunar Twins [32] improves lunar knowledge retrieval and question answering by integrating a cross-topic corpus with RAG. For mission operations, KGs-enhanced LLMs have been developed to diagnose astronaut task states [31]. For unstructured scientific literature, recent studies use LLMs to extract edges and entities [24, 34], enabling the construction of resource databases. To assess task-oriented reasoning for lunar exploration, Lunar-Bench [33] has been introduced together with the Lunar-R1 reasoning model. In parallel, SpaceQA [10] provides a benchmark for open-domain question answering in space missions. For multimodality, Space-LLaVa [8] leverages synthetic datasets, including AI4Mars [30], for supervised fine-tuning, supporting joint image-text reasoning for perception and autonomous planning. Beyond these efforts, datasets such as Mars-Bench [26], LuSNAR [20], LunarRock3D [2] and LunarStereo [11] further broaden capabilities for Space understanding and analysis.

3 Lunar-KAG

3.1 Hypergraph Construction

Scientific corpora inherently contain n -ary relational dependencies. To capture these multi-entity facts, we formulate the unstructured text as a knowledge hypergraph $\mathcal{G}_H = (V, E_H)$.

3.1.1 Entity Node. Each node $v \in V$ denotes an independent scientific concept. We prompt DeepSeek-R1 [12] to extract the discrete entity set V from raw documents. An encoding layer then maps each node v into a continuous vector $\mathbf{h}_v \in \mathbb{R}^d$, yielding the entity feature matrix $\mathbf{H}_V \in \mathbb{R}^{|V| \times d}$.

3.1.2 Hyperedge. Based on the extracted entities, we further parse the unstructured text to identify n -ary relational dependencies, instantiating each relation as a hyperedge e . The textual description of this event is then encoded into a continuous semantic vector $\mathbf{h}_e \in \mathbb{R}^d$, yielding the hyperedge feature matrix $\mathbf{H}_E \in \mathbb{R}^{|E_H| \times d}$.

3.1.3 Graph Representation. We formulate the topology of the $\mathcal{G}_H$ using an incidence matrix $\mathbf{A} \in \mathbb{R}^{|V| \times |E_H|}$, defined as:

$$A_{v,e} = \begin{cases} 1, & \text{if } v \in e \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The incidence matrix $\mathbf{A}$, combined with the semantic matrices $\mathbf{H}_V$ and $\mathbf{H}_E$, completely decouples the hypergraph into orthogonal structural and semantic spaces. These three matrices constitute the unified input for the downstream retrieval layer.

3.2 Dual-Path Retrieval

An encoder maps a given query q into a dense representation $\mathbf{h}_q \in \mathbb{R}^d$. To capture both independent concepts and higher-order relations, a dual-path mechanism performs parallel dense retrieval across the decoupled entity and hyperedge semantic spaces.

3.2.1 Entity Path Retrieval. In the entity path, we calculate the cosine similarity $s_V(q, v)$ between the query vector $\mathbf{h}_q$ and each entity vector $\mathbf{h}_v \in \mathbf{H}_V$:

$$s_V(q, v) = \frac{\mathbf{h}_q \cdot \mathbf{h}_v}{\|\mathbf{h}_q\| \|\mathbf{h}_v\|} \quad (2)$$

An absolute similarity threshold τ_V is applied to filter the entity seed set $\mathcal{R}_V(q)$:

$$\mathcal{R}_V(q) = \{v \in V \mid s_V(q, v) \geq \tau_V\} \quad (3)$$

3.2.2 Hyperedge Path Retrieval. Concurrently, the hyperedge path evaluates the similarity $s_H(q, e)$ between $\mathbf{h}_q$ and each hyperedge vector $\mathbf{h}_e \in \mathbf{H}_E$. A Top- k selection isolates the most relevant relational dependencies to formulate the hyperedge seed set $\mathcal{R}_H(q)$:

$$\mathcal{R}_H(q) = \arg \max_{E' \subseteq E_H, |E'|=k} \sum_{e \in E'} s_H(q, e) \quad (4)$$

The retrieved discrete sets $\mathcal{R}_V(q)$ and $\mathcal{R}_H(q)$ serve as the structural anchors for the subsequent subgraph expansion phase.

3.3 Hypergraph-Guided Generation

Since dense retrieval yields isolated entities and hyperedges, we perform a 1-hop expansion via the incidence matrix $\mathbf{A}$ to recover the local topological context. Specifically, we extract the incident hyperedges $\mathcal{F}_E$ for the seed entities, and the constituent entities $\mathcal{F}_V$ for the seed hyperedges:

$$\mathcal{F}_E = \{e \in E_H \mid \exists v \in \mathcal{R}_V(q), \mathbf{A}_{v,e} = 1\} \quad (5)$$

$$\mathcal{F}_V = \{v \in V \mid \exists e \in \mathcal{R}_H(q), \mathbf{A}_{v,e} = 1\} \quad (6)$$

These expansions induce a sub-hypergraph $\mathcal{G}_{\text{sub}} = (V_{\text{sub}}, E_{\text{sub}})$, where the node and hyperedge sets are defined as:

$$V_{\text{sub}} = \mathcal{R}_V(q) \cup \mathcal{F}_V, \quad E_{\text{sub}} = \mathcal{R}_H(q) \cup \mathcal{F}_E \quad (7)$$

Finally, we serialize $\mathcal{G}_{\text{sub}}$ into a textual prompt, concatenating it with query q to guide the LLMs generation process.

4 Experiments

To evaluate the effectiveness of Lunar-KAG in domain-specific long-document question answering, we design experiments to address the following research questions:

- **RQ1:** Does Lunar-KAG outperform state-of-the-art baselines in overall multi-hop question-answering performance?
- **RQ2:** How do individual structural components and retrieval thresholds affect the model’s efficacy?
- **RQ3:** How does Lunar-KAG perform in terms of computational efficiency and monetary cost during the knowledge construction and generation phases?

4.1 Experimental Settings

Datasets. We evaluate our framework on **SpaceQA** [10], **CosmoPaperQA** [34], and our proposed **LunarRQA**. SpaceQA, designed by the ESA, requires retrieving relevant passages from a vast aerospace technical corpus to answer 60 specific evaluation questions. CosmoPaperQA targets generative reasoning over long astrophysics publications, providing 105 expert-annotated QA pairs. To establish a domain-specific benchmark for lunar science, we construct LunarRQA from a corpus of 700 articles in top-tier journals like *Nature*. Yielding 317 expert-annotated QA pairs, LunarRQA was processed following established protocols [15, 35] and subjected to strict multi-round manual quality screening by domain experts.

Baselines. We compare our method with the following baselines: BM25RAG [27], DenseRAG [16], and Naive RAG [18] serve as standard chunk-based retrieval methods. For advanced architectures, Self-RAG [1] employs an adaptive retrieval and reflection mechanism, while RAPTOR [28] builds a recursive tree structure for hierarchical retrieval. In the graph-based domain, HippoRAG [14] integrates a neurobiological memory model for structured retrieval. GraphRAG [5] represents the foundational KGs-based framework, whereas LightRAG [13] and MiniRAG [6] streamline the graph construction process to meet industrial deployment requirements.

Metrics. We evaluate retrieval and generation performance independently. Retrieval efficacy is measured using Recall@{5, 10} and Mean Reciprocal Rank (MRR@10). For generative QA, we assess alignment via ROUGE-L. Furthermore, we employ GPT-5 [29] as an *LLM-as-a-Judge* to evaluate factual Faithfulness on a 0–5 scale.

Implementation Details. We use Qwen3-235B-A22B-Thinking-2507 with greedy decoding ($T = 0.0$) for generation and BAAI/bge-m3 [3] for dual-path dense retrieval encoding. Retrieval hyperparameters are set to $k = 24$ and entity cosine similarity threshold $\tau_V = 0.75$, with a maximum context constraint of 60 global evidence elements. All baselines are evaluated using identical embeddings over 512-token chunks with a 50-token overlap.

4.2 Overall Performance (RQ1)

Table 1 reports the retrieval and generation metrics averaged over five independent runs across the three benchmarks. Chunk-based methods (DenseRAG, RAPTOR) struggle with long-form scientific documents; for instance, DenseRAG achieves an MRR@10 of only 35.6% and a Faithfulness of 2.85 on SpaceQA. Graph-based baselines exhibit competitive performance, with HippoRAG and LightRAG alternately achieving sub-optimal results. In the retrieval phase, Lunar-KAG achieves an MRR@10 of 53.6% and a Recall@10 of 72.8% on LunarRQA, yielding absolute improvements of 4.2% and 4.3% over the strongest performing baseline, LightRAG. On SpaceQA, Lunar-KAG records an MRR@10 of 54.3% and a Recall@5 of 63.8%, surpassing LightRAG by 3.2% and 3.5%, respectively. On CosmoPaperQA, the MRR@10 reaches 51.2%, outperforming LightRAG by 3.4%. For the open-domain generation phase, Lunar-KAG records the highest ROUGE-L scores and explicitly maximizes factual correctness. On LunarRQA, Lunar-KAG improves the ROUGE-L score to 50.8%, a 3.7% absolute increase over LightRAG. Evaluated on a 0–5 scale via the GPT-5, Lunar-KAG improves Faithfulness from 3.82 to 4.15 on CosmoPaperQA, and from 4.02 to 4.38 on LunarRQA. On SpaceQA, the Faithfulness score reaches a peak of 4.45.

Table 1: Overall retrieval and generation performance across three benchmarks. All reported metrics are averaged over 5 independent runs. The best results are in bold with purple, and the second-best are underlined with yellow.

Method	SpaceQA					CosmoPaperQA					LunarRQA				
	MRR@10	R@5	R@10	R-L	Faith.	MRR@10	R@5	R@10	R-L	Faith.	MRR@10	R@5	R@10	R-L	Faith.
BM25RAG	22.4	31.5	40.2	28.4	2.15	18.3	25.6	33.5	24.1	1.85	20.1	28.4	37.6	25.8	2.05
DenseRAG	35.6	42.1	51.5	36.5	2.85	31.2	38.5	48.2	32.8	2.45	33.5	40.2	49.5	34.2	2.65
RAPTOR	41.2	48.3	58.2	40.2	3.15	36.8	44.2	53.6	37.5	2.75	38.9	46.5	56.4	38.9	2.95
Self-RAG	43.5	51.2	61.8	42.1	3.35	39.4	47.6	57.1	40.2	2.95	41.2	49.8	59.3	41.5	3.12
GraphRAG	46.8	55.4	65.1	44.5	3.55	42.1	50.3	60.5	42.8	3.15	44.5	52.6	62.8	43.6	3.35
HippoRAG	49.2	<u>60.3</u>	<u>70.1</u>	46.8	<u>4.12</u>	<u>47.8</u>	54.8	64.2	<u>46.7</u>	3.75	<u>49.4</u>	57.1	<u>68.5</u>	45.8	<u>4.02</u>
LightRAG	<u>51.1</u>	59.1	69.2	<u>48.2</u>	4.05	46.2	<u>56.4</u>	<u>66.1</u>	45.3	<u>3.82</u>	48.7	<u>58.3</u>	67.2	<u>47.1</u>	3.95
MiniRAG	47.3	57.6	67.5	45.2	3.65	44.8	53.5	62.4	43.9	3.25	46.2	54.1	64.1	44.8	3.45
Lunar-KAG	54.3	63.8	73.2	51.4	4.45	51.2	59.7	69.4	49.5	4.15	53.6	62.5	72.8	50.8	4.38

4.3 Ablation Study (RQ2)

To evaluate the contribution of each component, we conduct an ablation study on the LunarRQA with the following variants. Specifically, **w/o Entity** and **w/o Hyperedge** disable the entity and hyperedge retrieval paths, restricting dense matching exclusively to the hyperedge space H_E and the discrete entity space H_V , respectively. **w/ Standard KG** replaces the knowledge hypergraph $\mathcal{G}_H$ with a triplet-based knowledge graph, projecting n -ary events into binary edges. Finally, **w/o Expansion** removes the 1-hop topological expansion governed by the incidence matrix A , directly inputting the topologically disjoint seed sets $\mathcal{R}_V(q)$ and $\mathcal{R}_H(q)$ into the LLMs.

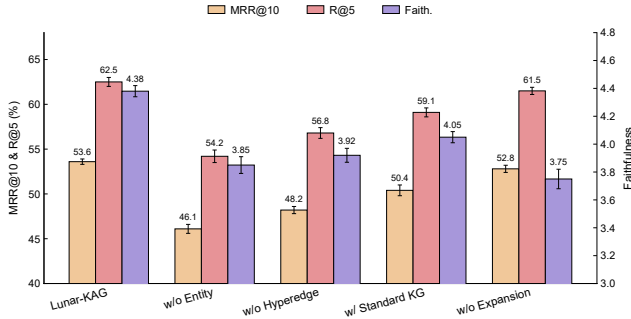
**Figure 2: Ablation study of Lunar-KAG on the LunarRQA.**

Figure 2 reports the performance of the ablated variants on the LunarRQA dataset. Removing the entity path (w/o Entity) and the hyperedge path (w/o Hyperedge) decreases MRR@10 by 7.5% and 5.4%, respectively. The discrete entity space H_V and the relational hyperedge space H_E provide distinct semantic features for dense retrieval. Replacing the hypergraph with a binary graph (w/ Standard KG) decreases MRR@10 by 3.2% and Faithfulness by 0.33. Projecting n -ary scientific events into triplets alters the topological structure of the matching space. Disabling subgraph expansion (w/o Expansion) results in a 0.63 decrease in Faithfulness. The induced sub-hypergraph $\mathcal{K}_H$ provides topological connections for the isolated retrieval seeds. Bypassing this expansion reduces the factual alignment of LLMs during multi-hop generation.

4.4 Efficiency and Cost Analysis (RQ3)

Table 2 reports the time and monetary costs of the evaluated methods during the knowledge construction and generation phases. The metrics include Time per 1k Tokens (TP1kT), Cost per 1k Tokens (CP1kT), Time Per Query (TPQ), and Cost per 1k Queries (CP1kQ). In the construction phase, Lunar-KAG records a TP1kT of 3.084s and a CP1kT of \$0.0063. This positions the construction overhead between HippoRAG2 (2.758s, \$0.0056) and GraphRAG (9.272s, \$0.0058). In the generation phase, Lunar-KAG requires 0.256s per query with a CP1kQ of \$3.184. While this incurs higher overhead than StandardRAG (0.147s, \$1.016) and GraphRAG (0.221s, \$1.836), Lunar-KAG reduces generation latency and monetary cost compared to PathRAG (0.436s, \$3.496) and LightRAG (0.359s, \$3.359).

Table 2: Time and cost consumption comparisons.

Method	Construction		Generation	
	TP1kT (s)	CP1kT (\$)	TPQ (s)	CP1kQ (\$)
Direct	0	0	0.131	0.059
Naive RAG	0	0	0.147	1.016
GraphRAG	9.272	0.0058	0.221	1.836
LightRAG	5.168	0.0081	0.359	3.359
PathRAG	5.168	0.0081	0.436	3.496
HippoRAG	2.758	0.0056	0.240	3.438
Lunar-KAG	3.084	0.0063	0.256	3.184

5 Conclusion

This paper presents Lunar-KAG, a RAG framework for scientific discovery in lunar exploration. The framework formalizes unstructured scientific text into a knowledge hypergraph, mapping multi-entity events to hyperedges to preserve higher-order topologies. A dual-path retrieval module executes dense matching independently within the decoupled entity space and the hyperedge space. Empirical evaluations on SpaceQA, CosmoPaperQA, and LunarRQA demonstrate that Lunar-KAG improves retrieval MRR@10 and generation Faithfulness compared to graph-based baselines, while reducing token consumption and latency relative to standard graph traversal approaches. Future work will extend the hypergraph structure to incorporate multimodal astronomical data.

Presenter Biography

Qianchen Xia is an Assistant Professor (AP) at the Future Laboratory, Tsinghua University, and leads the Human Factors Research Group. She received her PhD in Computer Science from Beihang University. Her research focuses on human–computer interaction for lunar exploration scenarios, and she has published at top-tier conferences including CVPR and ACL.

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